Template

Studentnames and studentnumbers here

2026-06-12

Set-up your environment

```
library(cbsodataR)
library(dplyr)
library(tidyr)
library(sf)
library(knitr)
library(ggplot2)
```

Title Page

Include your names Melvin Hofman Max Bergsma Gijs Stevens Hidde Voors Imad el Hilali

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Housing affordability has become an increasingly important social issue in the Netherlands over the past decade. House prices have risen substantially in many parts of the country, while income growth may not have kept pace. As a result, many households, particularly first-time buyers and younger generations, face greater difficulties entering the housing market. Rising housing costs can contribute to financial stress, wealth inequality, and reduced access to homeownership.

Understanding whether house prices have increased faster than incomes is therefore important for evaluating changes in housing affordability. If housing prices grow more rapidly than household incomes, purchasing a home becomes increasingly difficult for a large share of the population. This may have significant social and economic consequences, including increased inequality between homeowners and non-homeowners.

The main research question of this study is:

Have housing prices in the Netherlands increased faster than incomes over the past ten years?

To answer this question, the following sub-questions are examined: 1. How have house prices and average incomes developed between 2014 and 2024? 2. How has housing affordability changed during this period? 3. Are there significant differences in housing affordability between Dutch provinces? 4. Did the COVID-19 pandemic influence the relationship between house prices and income?

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
# data_download <- cbs_get_data("70072NED")

# Add region label to the dataset
# raw_dataset <- data_download %>% cbs_add_label_columns()
# write.csv(raw_dataset, "data/raw_dataset.csv")

# Load saved dataset
raw_dataset = read.csv("data/raw_dataset.csv")
```

2.2 Provide a short summary of the dataset(s)

```
huizenprijzen_provincie = raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2023,
    startsWith(RegioS, "PV")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    GemiddeldeWOZWaardeVanWoningen_98
  )
write.csv(huizenprijzen_provincie, "data/huizenprijzen_provincie.csv")

kable(huizenprijzen_provincie)
```

Year	RegioS	RegioS_label	GemiddeldeWOZWaardeVanWoningen_98
2023	PV20	Groningen (PV)	268
2023	PV21	Fryslân (PV)	277
2023	PV22	Drenthe (PV)	294
2023	PV23	Overijssel (PV)	325
2023	PV24	Flevoland (PV)	347
2023	PV25	Gelderland (PV)	364
2023	PV26	Utrecht (PV)	447
2023	PV27	Noord-Holland (PV)	460
2023	PV28	Zuid-Holland (PV)	359
2023	PV29	Zeeland (PV)	281
2023	PV30	Noord-Brabant (PV)	370
2023	PV31	Limburg (PV)	288

```
inkomen_provincie <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
```

```r
  ) %>%
  filter(
    Year == 2023,
    startsWith(RegioS, "PV")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    BronInkomenAlsWerknemer_141
  )
write.csv(inkomen_provincie, "data/inkomen_provincie.csv")

kable(inkomen_provincie)
```

Year	RegioS	RegioS_label	BronInkomenAlsWerknemer_141
2023	PV20	Groningen (PV)	37.5
2023	PV21	Fryslân (PV)	37.9
2023	PV22	Drenthe (PV)	39.7
2023	PV23	Overijssel (PV)	39.5
2023	PV24	Flevoland (PV)	40.4
2023	PV25	Gelderland (PV)	40.6
2023	PV26	Utrecht (PV)	44.4
2023	PV27	Noord-Holland (PV)	44.7
2023	PV28	Zuid-Holland (PV)	42.4
2023	PV29	Zeeland (PV)	40.3
2023	PV30	Noord-Brabant (PV)	41.8
2023	PV31	Limburg (PV)	39.9

```r
provincie_samengevoegd = huizenprijzen_provincie %>%
  inner_join(
    inkomen_provincie,
    by = c("Year", "RegioS", "RegioS_label")
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  )  %>%
  mutate(statcode = RegioS)
head(provincie_samengevoegd)
```

```
##   Year RegioS    RegioS_label GemiddeldeWOZ InkomenWerknemer statcode
## 1 2023 PV20    Groningen (PV)           268             37.5     PV20
## 2 2023 PV21      Fryslân (PV)           277             37.9     PV21
## 3 2023 PV22      Drenthe (PV)           294             39.7     PV22
## 4 2023 PV23   Overijssel (PV)           325             39.5     PV23
## 5 2023 PV24    Flevoland (PV)           347             40.4     PV24
## 6 2023 PV25   Gelderland (PV)           364             40.6     PV25
```

```r
provincie_summary = provincie_samengevoegd %>%
  summarise(
    Mean_WOZ = mean(GemiddeldeWOZ),
    Min_WOZ = min(GemiddeldeWOZ),
```

```
    Max_WOZ = max(GemiddeldeWOZ),
    SD_WOZ = sd(GemiddeldeWOZ),

    Mean_Income = mean(InkomenWerknemer),
    Min_Income = min(InkomenWerknemer),
    Max_Income = max(InkomenWerknemer),
    SD_Income = sd(InkomenWerknemer),
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = c("Statistic", "Variable"),
    names_sep = "_",
    values_to = "Value"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = Value
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ = "Average WOZ Value",
      Income = "Employee Income",
      Affordability = "WOZ-to-Income Ratio"
    )
  ) %>%
  select(Variable, Mean, Min, Max, SD)

kable(
  provincie_summary,
  caption = "Descriptive statistics for home value in provinces in the Netherlands, 2023",
  digits = 2
)
```

Table 3: Descriptive statistics for home value in provinces in the Netherlands, 2023

Variable	Mean	Min	Max	SD
Average WOZ Value	340.00	268.0	460.0	64.25
Employee Income	40.76	37.5	44.7	2.24

```
provincie_samengevoegd <- provincie_samengevoegd %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )
provincie_samengevoegd
```

```
##    Year RegioS       RegioS_label GemiddeldeWOZ InkomenWerknemer statcode
## 1  2023 PV20       Groningen (PV)           268             37.5     PV20
## 2  2023 PV21         Fryslân (PV)           277             37.9     PV21
## 3  2023 PV22         Drenthe (PV)           294             39.7     PV22
## 4  2023 PV23      Overijssel (PV)           325             39.5     PV23
## 5  2023 PV24       Flevoland (PV)           347             40.4     PV24
```

```
## 6  2023 PV25      Gelderland (PV)           364            40.6   PV25
## 7  2023 PV26         Utrecht (PV)           447            44.4   PV26
## 8  2023 PV27   Noord-Holland (PV)           460            44.7   PV27
## 9  2023 PV28    Zuid-Holland (PV)           359            42.4   PV28
## 10 2023 PV29         Zeeland (PV)           281            40.3   PV29
## 11 2023 PV30   Noord-Brabant (PV)           370            41.8   PV30
## 12 2023 PV31         Limburg (PV)           288            39.9   PV31
##    AffordabilityRatio
## 1            7.146667
## 2            7.308707
## 3            7.405542
## 4            8.227848
## 5            8.589109
## 6            8.965517
## 7           10.067568
## 8           10.290828
## 9            8.466981
## 10           6.972705
## 11           8.851675
## 12           7.218045
```

```r
# 2e Nieuwe variable

write.csv(provincie_samengevoegd, "data/provincie_samengevoegd.csv")
```

```r
landelijk <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year >= 2015,
    Year <= 2024,
    startsWith(RegioS, "NL")
  ) %>%
  select(
    Year,
    RegioS_label,
    BronInkomenAlsWerknemer_141,
    GemiddeldeWOZWaardeVanWoningen_98
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer,
    WOZ_growth_yoy = (GemiddeldeWOZ / lag(GemiddeldeWOZ) - 1) * 100,
    Income_growth_yoy = (InkomenWerknemer / lag(InkomenWerknemer) - 1) * 100,
    GrowthGap_yoy = WOZ_growth_yoy - Income_growth_yoy
  )

write.csv(landelijk, "data/landelijk.csv")

landelijk
```

```
##    Year RegioS_label InkomenWerknemer GemiddeldeWOZ AffordabilityRatio
## 1  2015    Nederland             30.3           206           6.798680
## 2  2016    Nederland             31.4           208           6.624204
## 3  2017    Nederland             32.0           217           6.781250
## 4  2018    Nederland             32.3           230           7.120743
## 5  2019    Nederland             33.7           250           7.418398
## 6  2020    Nederland             35.3           271           7.677054
## 7  2021    Nederland             37.1           290           7.816712
## 8  2022    Nederland             39.0           317           8.128205
## 9  2023    Nederland             41.8           368           8.803828
## 10 2024    Nederland             44.7           378           8.456376
##    WOZ_growth_yoy Income_growth_yoy GrowthGap_yoy
## 1              NA                NA            NA
## 2       0.9708738          3.630363     -2.659489
## 3       4.3269231          1.910828      2.416095
## 4       5.9907834          0.937500      5.053283
## 5       8.6956522          4.334365      4.361287
## 6       8.4000000          4.747774      3.652226
## 7       7.0110701          5.099150      1.911920
## 8       9.3103448          5.121294      4.189051
## 9      16.0883281          7.179487      8.908841
## 10      2.7173913          6.937799     -4.220408
```

```r
provincie_map <- cbs_get_sf("provincie", 2023, verbose = TRUE)
```

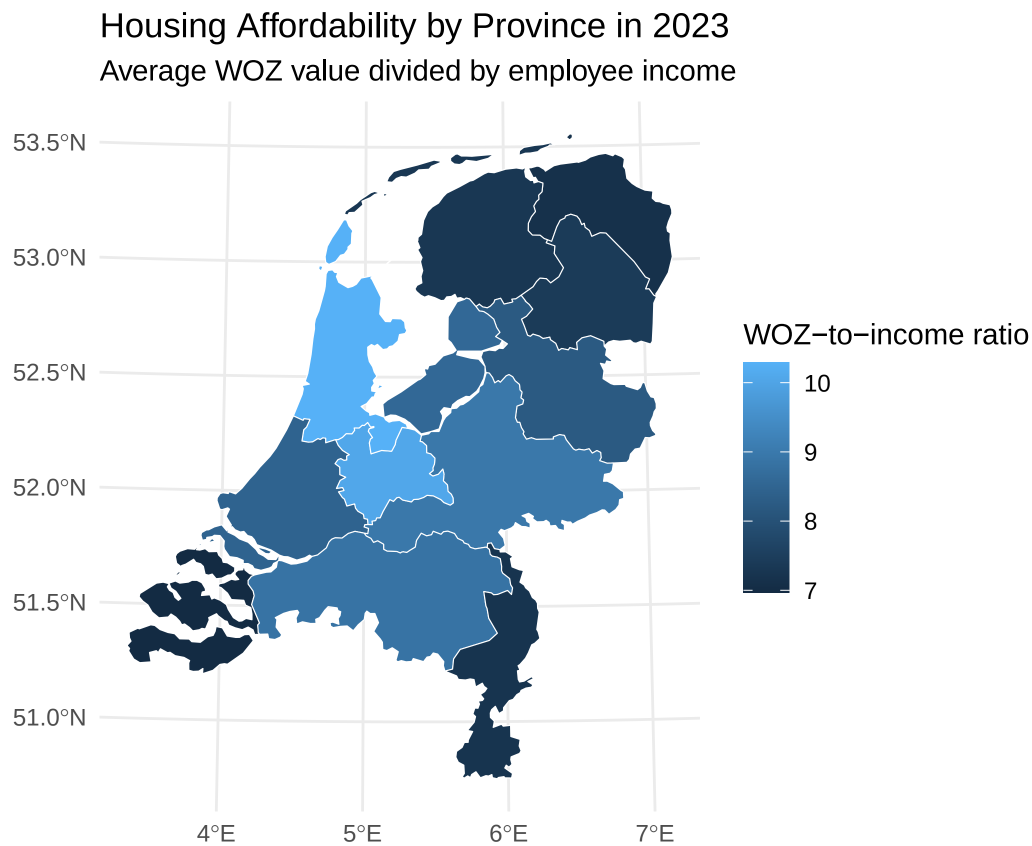
```
## Note that `cbs_add_statcode` can be used for connecting the map to the data.
```

```
## Retrieving list of geojson files: 'https://cartomap.github.io/nl/index.csv'
```

```
## Reading layer `provincie_2023' from data source
##   `https://cartomap.github.io/nl/rd/provincie_2023.geojson' using driver `GeoJSON'
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 13565 ymin: 306960 xmax: 277529 ymax: 619172
## Projected CRS: Amersfoort / RD New
```

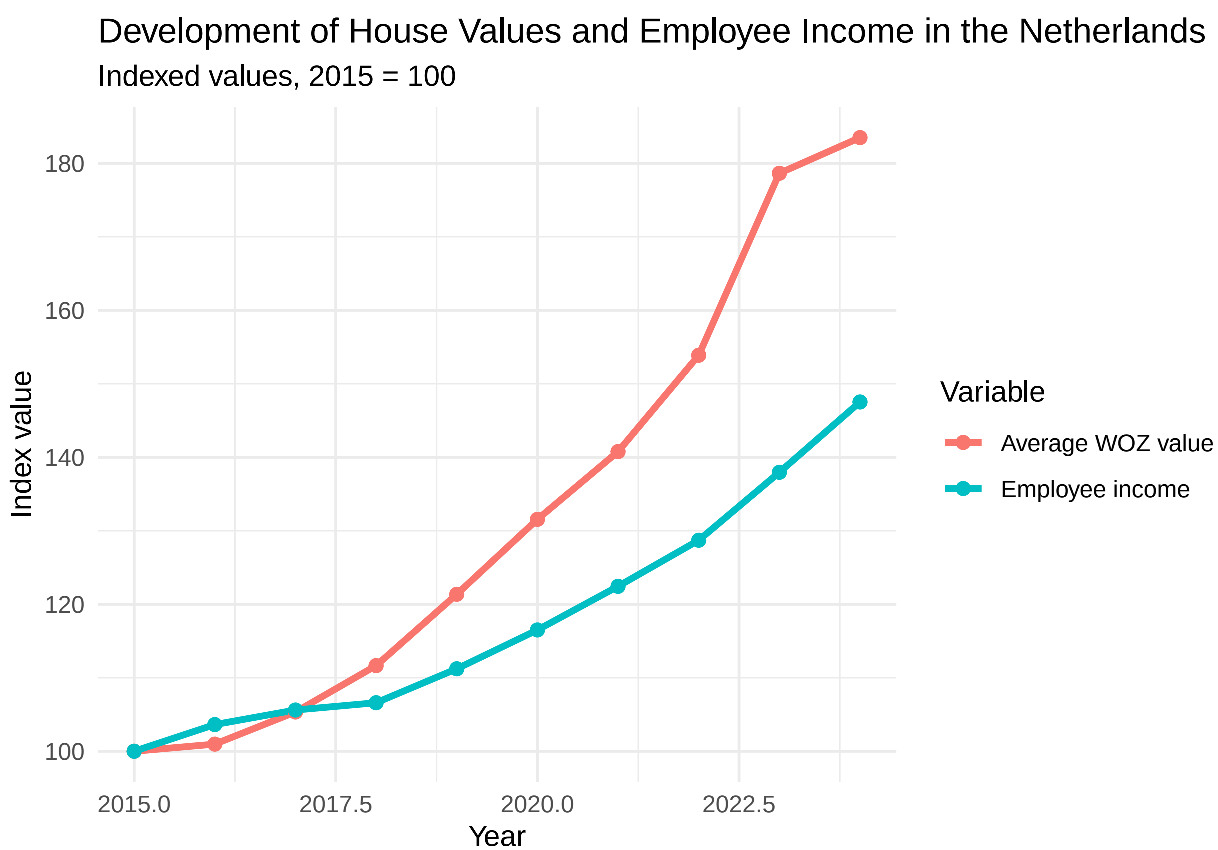
```r
provincie_samengevoegd <- provincie_samengevoegd %>%
  mutate(
    statcode = trimws(RegioS),
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )

provincie_map <- provincie_map %>%
  mutate(
    statcode = trimws(statcode)
  )

provincie_map_data <- provincie_map %>%
  left_join(
    provincie_samengevoegd,
    by = "statcode"
  )

ggplot(provincie_map_data) +
  geom_sf(aes(fill = AffordabilityRatio), color = "white") +
```

```r
  labs(
    title = "Housing Affordability by Province in 2023",
    subtitle = "Average WOZ value divided by employee income",
    fill = "WOZ-to-income ratio"
  ) +
  theme_minimal()
```



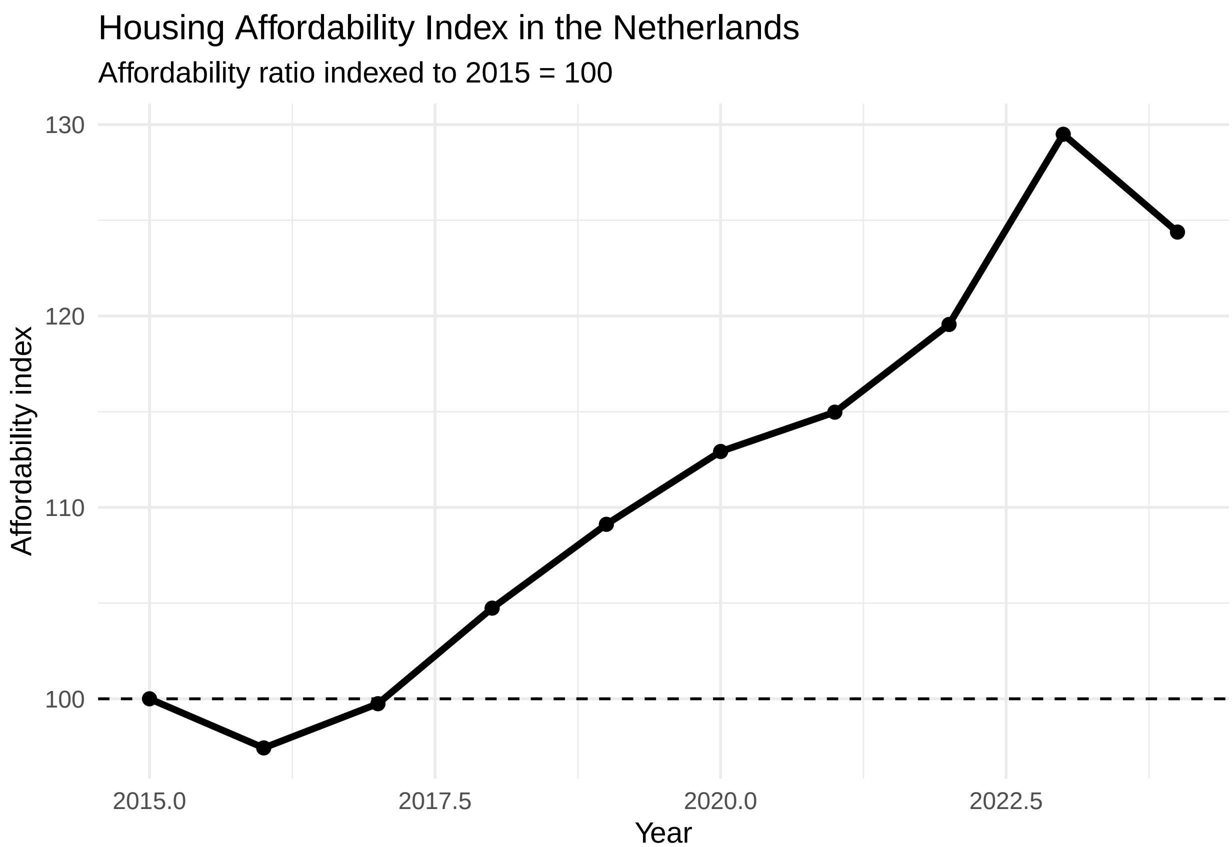

```r
# Deze plot gemaakt met chat

landelijk_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    WOZ_index = GemiddeldeWOZ / GemiddeldeWOZ[Year == 2015] * 100,
    Inkomen_index = InkomenWerknemer / InkomenWerknemer[Year == 2015] * 100
  ) %>%
  select(Year, WOZ_index, Inkomen_index) %>%
  pivot_longer(
    cols = c(WOZ_index, Inkomen_index),
    names_to = "Variable",
    values_to = "Index"
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ_index = "Average WOZ value",
      Inkomen_index = "Employee income"
```

```
    )
  )

ggplot(landelijk_index, aes(x = Year, y = Index, color = Variable)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Development of House Values and Employee Income in the Netherlands",
    subtitle = "Indexed values, 2015 = 100",
    x = "Year",
    y = "Index value",
    color = "Variable"
  ) +
  theme_minimal()
```




```
landelijk_affordability_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    AffordabilityIndex = AffordabilityRatio / AffordabilityRatio[Year == 2015] * 100
  )

ggplot(landelijk_affordability_index, aes(x = Year, y = AffordabilityIndex)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  geom_hline(yintercept = 100, linetype = "dashed") +
  labs(
```

```
    title = "Housing Affordability Index in the Netherlands",
    subtitle = "Affordability ratio indexed to 2015 = 100",
    x = "Year",
    y = "Affordability index"
  ) +
  theme_minimal()
```

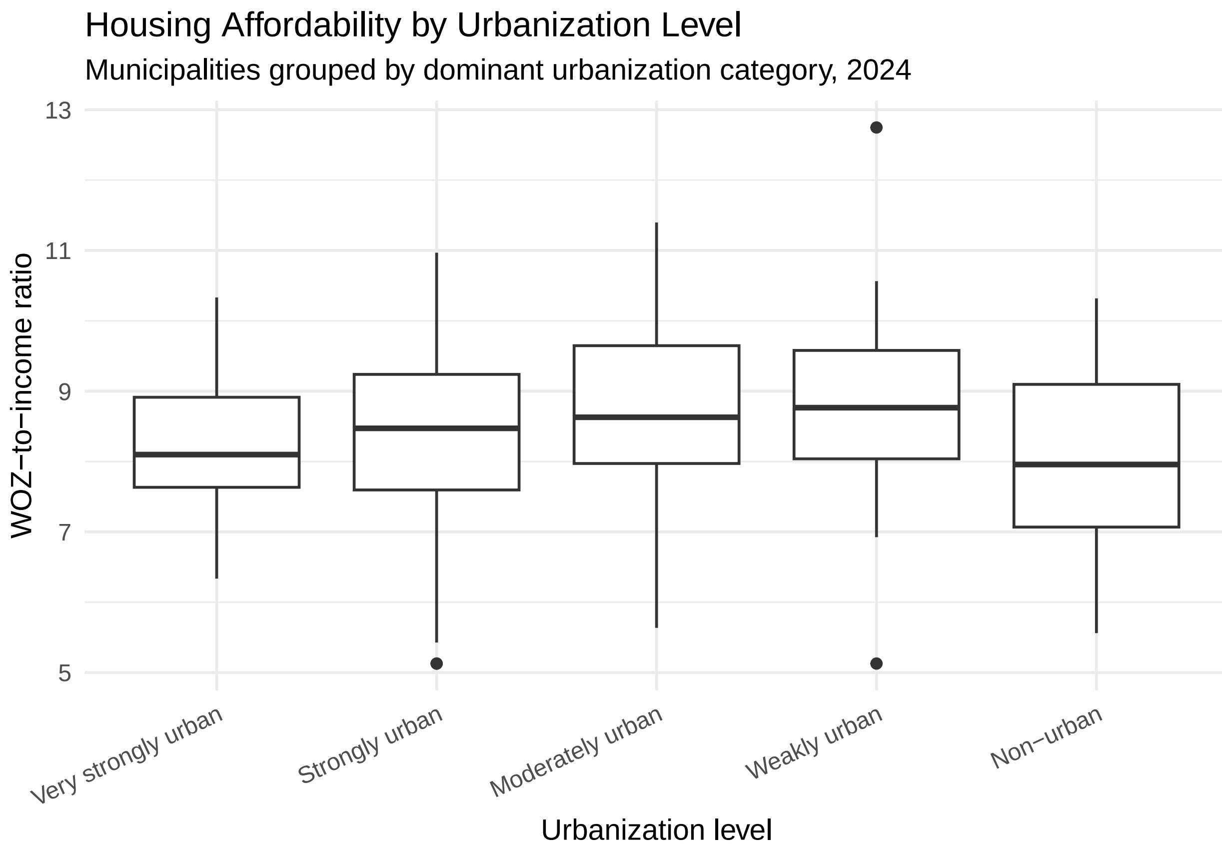



```
urban_affordability <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2024,
    startsWith(RegioS, "GM")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    GemiddeldeWOZWaardeVanWoningen_98,
    BronInkomenAlsWerknemer_141,
    ZeerSterkStedelijk_52,
    SterkStedelijk_53,
    MatigStedelijk_54,
    WeinigStedelijk_55,
    NietStedelijk_56
```

```r
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  ) %>%
  pivot_longer(
    cols = c(
      ZeerSterkStedelijk_52,
      SterkStedelijk_53,
      MatigStedelijk_54,
      WeinigStedelijk_55,
      NietStedelijk_56
    ),
    names_to = "UrbanizationType",
    values_to = "UrbanizationValue"
  ) %>%
  group_by(RegioS, RegioS_label, Year) %>%
  slice_max(UrbanizationValue, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  mutate(
    UrbanizationType = recode(
      UrbanizationType,
      ZeerSterkStedelijk_52 = "Very strongly urban",
      SterkStedelijk_53 = "Strongly urban",
      MatigStedelijk_54 = "Moderately urban",
      WeinigStedelijk_55 = "Weakly urban",
      NietStedelijk_56 = "Non-urban"
    ),
    UrbanizationType = factor(
      UrbanizationType,
      levels = c(
        "Very strongly urban",
        "Strongly urban",
        "Moderately urban",
        "Weakly urban",
        "Non-urban"
      )
    )
  )

ggplot(urban_affordability, aes(x = UrbanizationType, y = AffordabilityRatio)) +
  geom_boxplot() +
  labs(
    title = "Housing Affordability by Urbanization Level",
    subtitle = "Municipalities grouped by dominant urbanization category, 2024",
    x = "Urbanization level",
    y = "WOZ-to-income ratio"
  ) +
  theme_minimal() +
  theme(
```

```
    axis.text.x = element_text(angle = 25, hjust = 1)
  )
```

```
## Warning: Removed 408 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment…

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
#mean(dataset$percollege)
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

Variable 2

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.